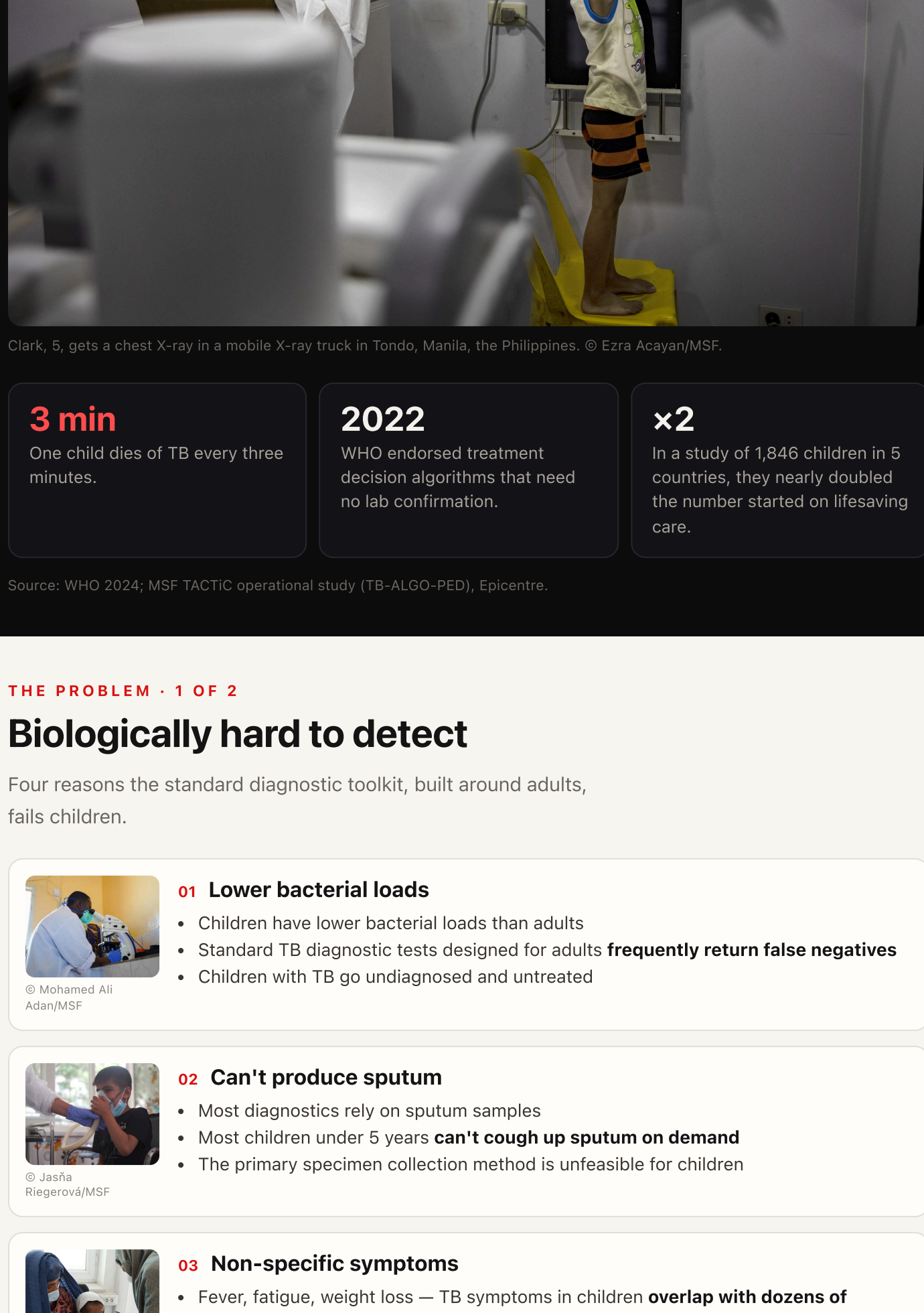


A child dies of TB every 3 minutes.

Almost none of them are diagnosed. A proven diagnostic solution exists, but it is not being used.

[MORE DIAGNOSIS](#) → [MORE TREATMENT](#) → [MORE LIVES SAVED](#)



Clark, 5, gets a chest X-ray in a mobile X-ray truck in Tondo, Manila, the Philippines. © Ezra Acayan/MSF.

3 min

One child dies of TB every three minutes.

2022

WHO endorsed treatment decision algorithms that need no lab confirmation.

x2

In a study of 1,846 children in 5 countries, they nearly doubled the number started on lifesaving care.

Source: WHO 2024; MSF TACTIC operational study (TB-ALGO-PED), Epicentre.

THE PROBLEM - 1 OF 2

Biologically hard to detect

Four reasons the standard diagnostic toolkit, built around adults, fails children.

- 01 Lower bacterial loads**

 - Children have lower bacterial loads than adults
 - Most diagnostics tests designed for adults **frequently return false negatives**
 - Children with TB go undiagnosed and untreated
- 02 Can't produce sputum**

 - Most diagnostics rely on sputum samples
 - Most children under 5 years **can't cough up sputum on demand**
 - The primary specimen collection method is unfeasible for children
- 03 Non-specific symptoms**

 - Fever, fatigue, weight loss — TB symptoms in children **overlap with dozens of common illnesses**
 - This causes under-diagnosis and dangerous delays
- 04 Extrapulmonary TB**

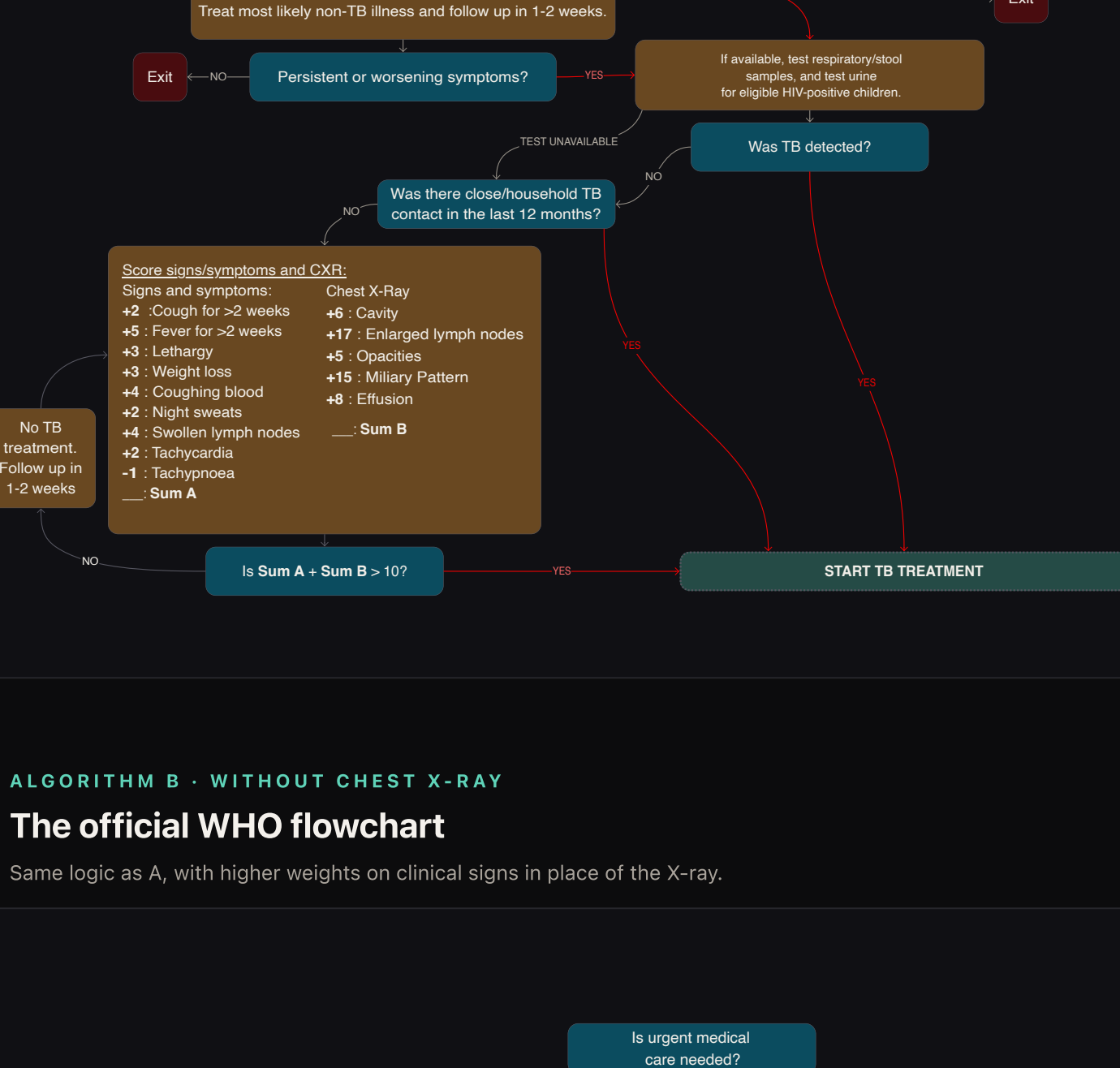
 - Children get TB in **lymph nodes, bones, brain and other non-lung sites** more frequently than adults
 - Sample collection and bacteriological confirmation become harder

THE PROBLEM - 2 OF 2

Children are politically *neglected*

Public health systems and biotechnology prioritise what is scalable, profitable and measurable, resulting in a self-reinforcing cycle that keeps children with TB sidelined.

THE VICIOUS CYCLE



What breaks the cycle: a diagnostic method that does not depend on adult-style laboratory confirmation. That method already exists and is endorsed by the WHO.

A PROVEN SOLUTION

WHO Treatment Decision Algorithms: *a diagnostic flowchart that saves lives*

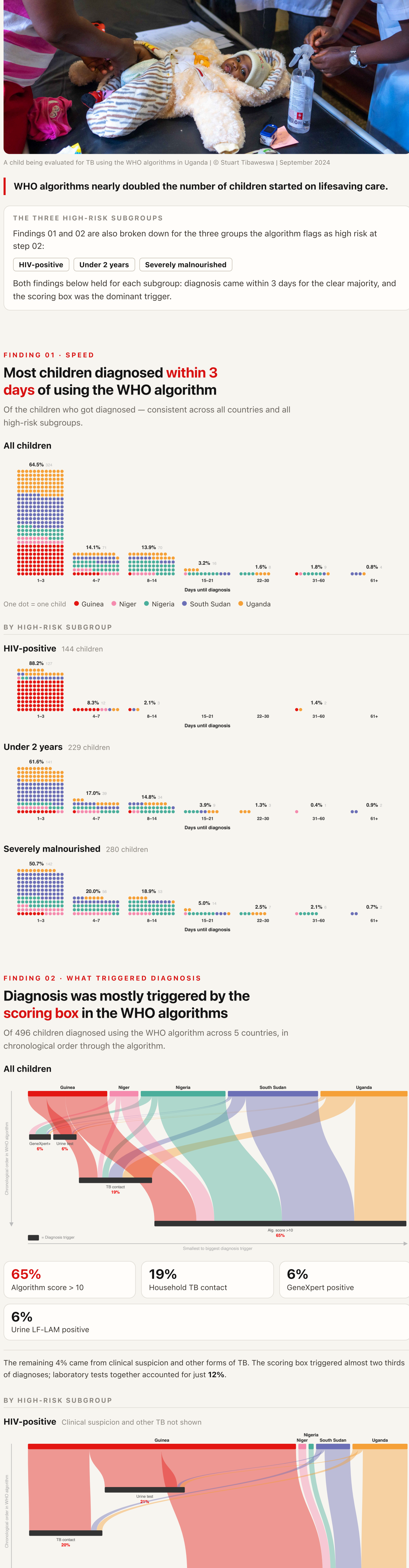
A scoring system that lets a clinician start TB treatment without laboratory confirmation.

- Identifies pulmonary TB in **children under 10 years** without advanced technology or specialists
 - Systematises **non-specific symptoms**, TB exposure, basic test results and chest X-ray results (when available) into a cohesive scoring system
 - Endorsed by the **World Health Organization** in 2022
- A score **above 10** triggers TB treatment — no lab confirmation required
- 🔄 **Two versions:** A for clinics with chest X-ray, B for those without — identical logic, different scoring weights
- 📊 Based on data pooled from **thousands of children** across multiple countries

ALGORITHM A • WITH CHEST X-RAY

The official WHO flowchart

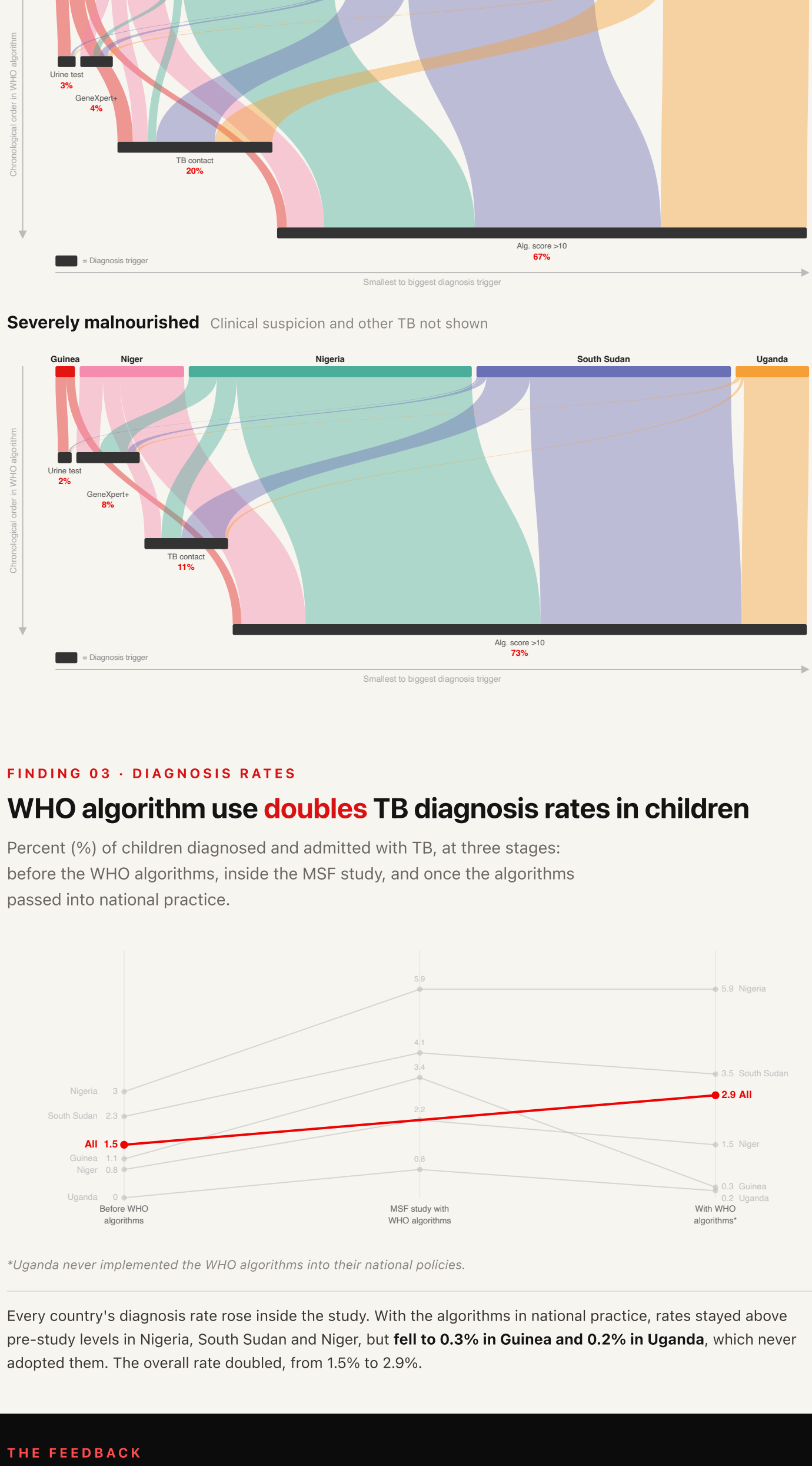
Vector diagram — zoom in to read the scoring box.



ALGORITHM B • WITHOUT CHEST X-RAY

The official WHO flowchart

Same logic as A, with higher weights on clinical signs in place of the X-ray.

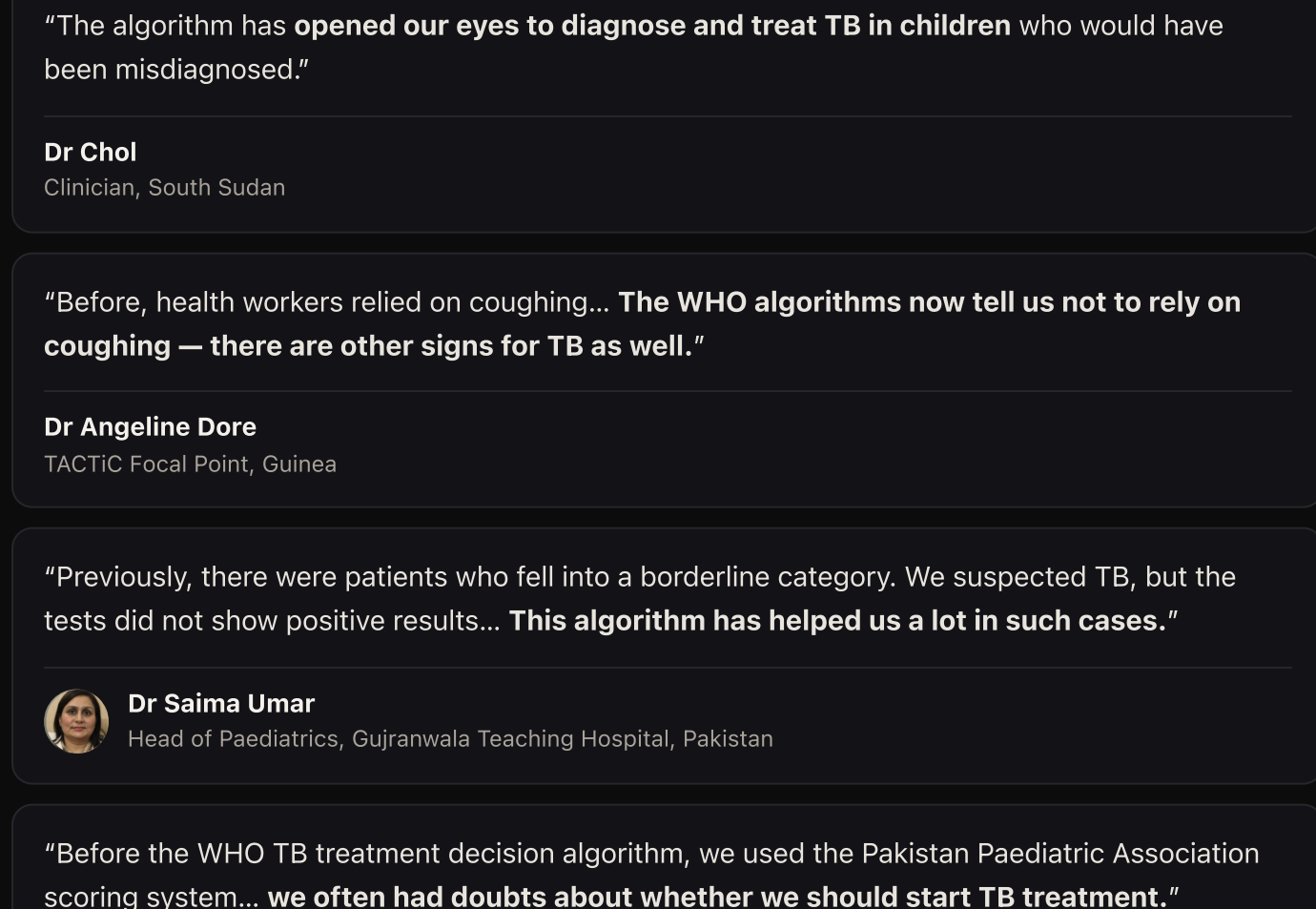


THE EVIDENCE

1,846 children. 5 countries. 1 result.

MSF's TACTIC initiative ran the **TB ALGO PED** study to test the diagnostic accuracy and implementability of the WHO algorithms. It enrolled **1,846** children under 10 years with symptoms consistent with TB.

[Guinea](#) [Niger](#) [Nigeria](#) [South Sudan](#) [Uganda](#)



A child being evaluated for TB using the WHO algorithms in Uganda | © Stuart Tibaweswa | September 2024

WHO algorithms nearly doubled the number of children started on lifesaving care.

THE THREE HIGH-RISK SUBGROUPS

Findings 01 and 02 are also broken down for the three groups the algorithm flags as high risk at step 02:

[HIV-positive](#) [Under 2 years](#) [Severely malnourished](#)

Both findings below held for each subgroup: diagnosis came within 3 days for the clear majority, and the scoring box was the dominant trigger.

FINDING 01 • SPEED

Most children diagnosed **within 3 days** of using the WHO algorithm

Of the children who got diagnosed — consistent across all countries and all high-risk subgroups.

All children

One dot = one child. ● Guinea ● Niger ● Nigeria ● South Sudan ● Uganda

BY HIGH-RISK SUBGROUP

HIV-positive 144 children

Under 2 years 229 children

Severely malnourished 260 children

FINDING 02 • WHAT TRIGGERED DIAGNOSIS

Diagnosis was mostly triggered by the **scoring box** in the WHO algorithms

Of 496 children diagnosed using the WHO algorithm across 5 countries, in chronological order through the algorithm.

All children

65% Algorithm score > 10

19% Household TB contact

6% GeneXpert positive

6% Urine LF-LAM positive

The remaining 4% came from clinical suspicion and other forms of TB. The scoring box triggered almost two thirds of diagnoses; laboratory tests together accounted for just 12%.

BY HIGH-RISK SUBGROUP

HIV-positive Clinical suspicion and other TB not shown

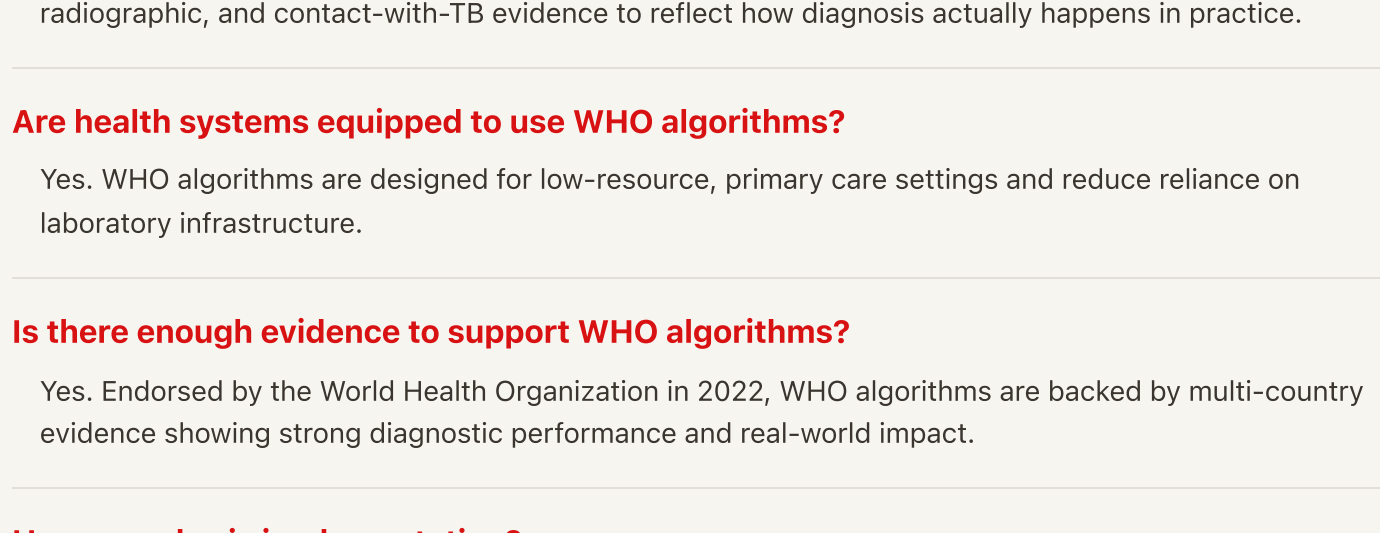
Under 2 years Clinical suspicion and other TB not shown

Severely malnourished Clinical suspicion and other TB not shown

FINDING 03 • DIAGNOSIS RATES

WHO algorithm use **doubles** TB diagnosis rates in children

Percent (%) of children diagnosed and admitted with TB, at three stages: before the WHO algorithms, inside the MSF study, and once the algorithms passed into national practice.



*Uganda never implemented the WHO algorithms into their national policies.

Every country's diagnosis rate rose inside the study. With the algorithms in national practice, rates stayed above pre-study levels in Nigeria, South Sudan and Niger, but **fell to 0.3% in Guinea and 0.2% in Uganda**, which never adopted them. The overall rate doubled, from 1.5% to 2.9%.

THE FEEDBACK

Clinicians say diagnosis with WHO algorithms is more *accessible, streamlined and effective*.

In their own words, from Guinea, Niger, Nigeria, South Sudan and Pakistan, and from MSF staff.

DIAGNOSTIC SHIFT

"The WHO algorithms have streamlined the diagnostic and treatment decision-making approach for TB in children — a process that had long remained challenging. They allow us improve the prognosis for children with TB by facilitating **early diagnosis**."

Dr Moussa Mamane Oumarou Farouk
TB Focal Point, Niger

"The algorithm has **opened our eyes to diagnose and treat TB** in children who would have been misdiagnosed."

Dr Chol
Clinician, South Sudan

"Before, health workers relied on coughing... **The WHO algorithms now tell us not to rely on coughing — there are other signs for TB as well.**"

Dr Angeline Dore
TACTIC Focal Point, Guinea

"Previously, there were patients who fell into a borderline category. We suspected TB, but the tests did not show positive results... **This algorithm has helped us a lot in such cases.**"

Dr Saima Umar
Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan

"Before the WHO TB treatment decision algorithm, we used the Pakistan Paediatric Association scoring system... **we often had doubts about whether we should start TB treatment.**"

Dr Saima Umar
Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan

IMPACT

"Thanks to these algorithms, we have **diagnosed tuberculosis earlier** and saved more children in health facilities with limited resources."

Dr Ibrahim Barry
Local Principal Investigator, Algo TB Ped study, Guinea

"Our findings prove that WHO's treatment decision algorithms, for which we do not need lab test results to initiate TB treatment in children, work in real-world settings and can potentially **save many more children's lives** if implemented."

Dr Helena Huerga
Principal Investigator, TACTIC study, MSF

"The WHO's recommendations serve as a **crucial tool to help health workers diagnose TB** in children in **all settings**, even when there is limited access to laboratory tests and X-ray..."

Dr Gabriella Ferlazzo
TB Medical Advisor, MSF Access Campaign

"It has also **significantly reduced delays in starting treatment** in children. TB can spread quickly and lead to serious infections. Any delay in starting treatment can worsen their condition."

Dr Saima Umar
Head of Paediatrics, Gujranwala Teaching Hospital, Pakistan

"Using the algorithm is **easy, feasible and acceptable** to follow in the Malakal Project."

Dr Lazro Fidelle Nyikayo
Study Coordinator, South Sudan

IMPLEMENTATION GAP

"The science is clear — what's now missing is the **political will to put it into practice.**"

Dr Helena Huerga
Principal Investigator, TACTIC study, MSF

"In an already underfunded TB response, children are **forced further to the back of the line** when services are disrupted by aid cuts, conflict, or displacement."

Dr Cathy Hewson
MSF Working Group Lead

"For algorithms to be sustained in our hospitals, **regular monitoring and follow-up is important.**"

Dr Mohammed Bashir Abdullahi
Medical Activity Manager, Nigeria

CALL TO ACTION

"We urge governments to adopt and put the WHO recommendations into practice... Governments must **leave no stone unturned to find every child with TB and treat them right away.**"

Dr Gabriella Ferlazzo
TB Medical Advisor, MSF Access Campaign

"The impact of implementing the WHO treatment decision algorithms in Niger has been a **catalyst for hope.**"

Dr Moussa Mamane Oumarou Farouk
TB Focal Point, Niger

ADDRESSING CONCERNS

Frequently asked questions

An MSF nurse palpates 7-year-old Vedant's lymph nodes in Mumbai, India. His sister has drug-resistant TB. © Prem Hessekamp.

How accurate are the WHO algorithms, and is there a risk of over-treatment?

False positives are preferred to false negatives — a deliberate design choice.

Sensitivity ~90.5%

Correctly identifies most children who do have TB — so fewer sick children are missed and go untreated.

Specificity ~93.8%

Some children who don't have TB will still be flagged — a deliberate tradeoff. **Missing a child in a disease this deadly costs more than over-treating.**

Aren't lab tests more reliable than the WHO algorithms?

Not in children. Current lab tests often fail to detect pediatric TB. WHO algorithms use clinical, radiographic, and contact-with-TB evidence to reflect how diagnosis actually happens in practice.

Are health systems equipped to use WHO algorithms?

Yes. WHO algorithms are designed for low-resource, primary care settings and reduce reliance on laboratory infrastructure.

Is there enough evidence to support WHO algorithms?

Yes. Endorsed by the World Health Organization in 2022, WHO algorithms are backed by multi-country evidence showing strong diagnostic performance and real-world impact.

How complex is implementation?

No special devices or new equipment are needed. Rolling them out requires staff training. WHO algorithm integration into existing workflows and anticipating drug needs for treatment.

However, the main barrier is policy inertia and incentive structures that favour lab-confirmed cases over clinically diagnosed children.

ACT NOW

The WHO algorithms are ready. **We know they work. Children are waiting.**

Vaishnavi, 7, has DR-TB and interacts with MSF nurse Prachi in Mumbai, India. February 2022. © Prem Hessekamp.

The WHO algorithms are validated in the field and can be integrated into National TB Programmes today.

WHERE TO GO NEXT

WHO guidelines - [tbksp.who.int/en/node/2032](#)

About TACTIC - [msf.org/tactic-tuberculosis-children](#)

About TB-ALGO-PED - [epicentre.msf.org/en/tb-algo-ped](#)

Contact - [tic.tb-children@paris.msf.org](#)

READ THE INTERACTIVE VERSION

[salkothari.github.io/tmsf-scrollytelling](#)

Médecins Sans Frontières - TACTIC - Epicentre

Data: MSF TACTIC operational study (TB-ALGO-PED), Epicentre

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